# Griffin Edwards

Ph.D. Candidate in Mathematics, Georgia Institute of Technology

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## Research Interests

Algebraic combinatorics, focusing on matroid theory, tropical geometry, algebraic geometry, and pastures.

## Education

**Georgia Institute of Technology, Atlanta, Georgia** *August 2022–May 2027 (expected)*
Doctor of Philosophy in Mathematics
Doctoral Minor in Higher Education and Tech to Teaching Certificate (in progress) – Center for Teaching and Learning
Dissertation (working title): *Dressians of Matroids through Pastures*
Advisor: Matthew Baker

**Brown University, Providence, Rhode Island** *September 2018–May 2022*
Bachelor of Science in Mathematics, with Honors

## Teaching

**Instructor of Record, Georgia Institute of Technology** *Fall 2026*
*MATH 1499: Support for Introductory Calculus*

- Taught a flipped, discussion-centered course for 15 students, using interactive readings, collaborative worksheets, and board presentations.
- Implemented an initial diagnostic assessment to identify common prerequisite gaps and incorporated targeted review into recurring class activities.
- Revised course materials to improve screen-reader accessibility and mathematical readability.

**Lecturer's Assistant, Georgia Institute of Technology** *Summer 2026*
*MATH 4107: Abstract Algebra I*

**Teaching Assistant, Georgia Institute of Technology** *Fall 2024–Spring 2026*

- *MATH 2603: Introduction to Discrete Mathematics* — Fall 2025 and Spring 2026
- *MATH 1552: Integral Calculus* — Spring 2025
- *MATH 1553: Introduction to Linear Algebra* — Fall 2024

**Undergraduate Teaching Assistant, Brown University** *2020–2021*
Teaching support for Honors Linear Algebra, Real Analysis, Differential Geometry, Linear Algebra, and Advanced Classical Mechanics.

**Tutor, Sheridan Center for Teaching and Learning, Brown University** *September 2019–May 2021*
Tutored students in mathematics, physics, and logic.

## Research Mentoring

**Research Mentor and Coordinator, Georgia Tech Mathematics REU** *Summer 2025*
Co-mentored Catie Lillja and Bora Olmez with Matthew Baker on *Tropical Elliptic Curve Group Law Algorithm.*

**Mentor, Georgia Tech Directed Reading Program** *Fall 2022-Spring 2026*
Two students a semester on topics such as Tutte Polynomial, Category Theory, Elliptic Curves, Group Theory in Crystal Structures

## Research & Scholarly Work

### Manuscripts in preparation

- *Foundations of Elliptic Matroids*, with Matthew Baker and Noah Solomon. Manuscript in preparation.
- *Foundations of Matroids, Part 3*, with Matthew Baker and Oliver Lorscheid. Manuscript in preparation.

### Software in development

- *Pastures*, a Macaulay2 package, with Justin Chen and Tianyi Zhang.
- *Rigidity*, a Macaulay2 package, with Kalina Mincheva, Daniel Irving Bernstein, and additional collaborators.

**Selected research contribution**

- Computationally verified the conjectured reduced-Dressian behavior of the elliptic matroids $T_n$ for $7 \leq n \leq 26$; this computation is acknowledged in Baker–Huh–Kummer–Lorscheid, *Lorentzian Polynomials and Matroids over Triangular Hyperfields* (2025).

## Presentations

*Title to be announced.* Tulane University, New Orleans, Louisiana. *November 18, 2026*

*Tropical Elliptic Curves and Matroids.* Poster presented at the ISM Discovery School *May 11–15, 2026*

*Macaulay2 and Matroids.* Computer Algebra Club/Seminar presentation, Georgia Institute of Technology. *November 6, 2025*

*Algebra from Projective Geometry.* Algebra Student Seminar, Georgia Institute of Technology. *November 17, 2023*

## Awards & Fellowships

MathLab Award for Top TA Helper, Georgia Tech School of Mathematics *2026*
Recognized for exceptional student support in the department's communal drop-in office-hour and study space.

GAANN Grant, U.S. Department of Education *January 2025–present*

C-PIES GradSTEP Grant, Georgia Tech College of Sciences *January–May 2025*
Funding for Georgia Tech REU and Summer Edge Program social and professional developement programing.

FESTA Fellowship, Georgia Institute of Technology *April 2024*
Awarded for superior academic and leadership skills

David L. Brown Fellowship Award, Georgia Institute of Technology *April 2023*

Research to Training Group, Georgia Tech and National Science Foundation *August 2022*

Undergraduate Teaching and Research Award, Brown University *Summer 2020*

## Conferences & Workshops

Future of Macaulay2, Georgia Tech; contributor to packages group *May 2026*

Montreal ISM Discovery School, LACIM; poster presenter *May, 2026*

Macaulay2 Workshop, Tulane University; contributor *April 2025*

Tropical Geometry: Moduli Spaces and Matroids, Goethe University Frankfurt *October 2024*

Macaulay2, Computational Algebraic Geometry and String Theory; contributor *May 2024*

Preliminary Arizona Winter School; participant *Fall 2023*

Arizona Winter School; participant *March 2023*

Connecticut Summer School in Number Theory; participant *June 2022*

## Service, Leadership, & Outreach

Volunteer and Problem Writer, High School Math Day, Georgia Institute of Technology *2025–2027*

College of Sciences Graduate Student Committee Member, Georgia Tech *2024–present*
With C-PIES, select speakers for the *Making Science Accessible Seminar Series*

Graduate Peer Mentor Coordinator, Georgia Institute of Technology *2023–2026*

President, Graduate AMS Chapter, Georgia Institute of Technology *2023–2026*

Teacher, Emory Math Circle *2022–2026*

Graduate Student Representative, Department of Mathematics, Georgia Tech *2023–2024*
Representing the interests of the graduate student in faculty meetings.

President, Mathematics Department Undergraduate Group, Brown University *2020–2022*

Founder and Coordinator, Presentation Theory, Brown University *Fall 2021*
Planned and moderated mathematics seminars by undergraduates for undergraduates.

Volunteer, Brown University Math Circle *Fall 2021–Spring 2022*

## Technical Skills and Interests

**Software and programming:** Macaulay2, Python, and LaTeX.
**Other interests:** Board games, climbing, cooking, and juggling.

Last updated September 2026.